Exercises 5

Exercise 1 (Quantifier Elimination in PA)**.** Apply quantifier elimination as seen in the Lectures to the following formulas:

1. $\exists x, y.\, 2x + 3y < 7 \land x < y$
2. $\exists x, y.\, 2x + 3y < 7 \land y < x$
3. $\exists x, y.\, 3x + 3y < 8 \land 8 < 3x + 2y$
4. $\exists x, y.\, x = 2y \land \exists z.\, x = 3z$

Solution:

1. $\exists x, y.\, 2x + 3y < 7 \land x < y$

$$\begin{aligned} & \exists y.\, 2x + 3y < 7 \land x < y \\ \equiv\ & \exists y.\, 3x < 3y \land 3y < 7 - 2x \\ \equiv\ & \exists y'.\, 3x < y' \land y' < 7 - 2x \land 3 \mid y' \\ \equiv\ & \bigvee_{i=1}^{3} 3x + i < 7 - 2x \land 3 \mid 3x + i \end{aligned}$$

We can then proceed with the outer quantifier:

$$\begin{aligned} & \exists x.\, \bigvee_{i=1}^{3} 3x + i < 7 - 2x \land 3 \mid 3x + i \\ \equiv\ & \bigvee_{i=1}^{3} \exists x.\, 3x + i < 7 - 2x \land 3 \mid 3x + i \\ \equiv\ & \bigvee_{i=1}^{3} \exists x.\, 5x < 7 - i \land 3 \mid 3x + i \\ \equiv\ & \bigvee_{i=1}^{3} \exists x.\, 15x < 21 - 3i \land 15 \mid 15x + 5i \\ \equiv\ & \bigvee_{i=1}^{3} \exists x'.\, x' < 21 - 3i \land 15 \mid x' + 5i \land 15 \mid x' \\ \equiv\ & \bigvee_{i=1}^{3} \bigvee_{j=1}^{15} 15 \mid j + 5i \land 15 \mid j \\ \equiv\ & \mathit{True} \end{aligned}$$

2. $\exists x, y.\, 2x + 3y < 7 \wedge y < x$

$$\begin{aligned}
& \exists y.\, 2x + 3y < 7 \wedge y < x \\
\equiv\ & \exists y.\, 3y < 7 - 2x \wedge 3y < 3x \\
\equiv\ & \exists y'.\, y' < 7 - 2x \wedge y' < 3x \wedge 3 \mid y' \\
\equiv\ & \bigvee_{i=1}^{3} 3 \mid i \\
\equiv\ & \mathit{True}
\end{aligned}$$

We then have:

$$\exists x.\, \mathit{True} \ \equiv\ \mathit{True}$$

3. $\exists x, y.\, 3x + 3y < 8 \wedge 8 < 3x + 2y$

$$\begin{aligned}
& \exists y.\, 3x + 3y < 8 \wedge 8 < 3x + 2y \\
\equiv\ & \exists y.\, 8 - 3x < 2y \wedge 3y < 8 - 3x \\
\equiv\ & \exists y.\, 24 - 9x < 6y \wedge 6y < 16 - 6x \\
\equiv\ & \exists y'.\, 24 - 9x < y' \wedge y' < 16 - 6x \wedge 6 \mid y' \\
\equiv\ & \bigvee_{i=1}^{6} 24 - 9x + i < 16 - 6x \wedge 6 \mid 24 - 9x + i
\end{aligned}$$

We then have:

$$\begin{aligned}
& \exists x.\, \bigvee_{i=1}^{6} 24 - 9x + i < 16 - 6x \wedge 6 \mid 24 - 9x + i \\
\equiv\ & \exists x.\, \bigvee_{i=1}^{6} 8 + i < 3x \wedge 6 \mid 24 - 9x + i \\
\equiv\ & \exists x.\, \bigvee_{i=1}^{6} 24 + 3i < 9x \wedge 6 \mid 24 - 9x + i \\
\equiv\ & \exists x'.\, \bigvee_{i=1}^{6} 24 + 3i < x' \wedge 6 \mid 24 - x' + i \wedge 9 \mid x' \\
\equiv\ & \bigvee_{i=1}^{6} \bigvee_{j=1}^{18} 6 \mid 24 - j + i \wedge 9 \mid j \\
\equiv\ & \mathit{True}
\end{aligned}$$

4. $\exists x, y.\, x = 2y \wedge \exists z.\, x = 3z$

$$
\begin{aligned}
& \exists x, y.\, x = 2y \wedge \exists z.\, x = 3z \\
\equiv\ & \exists x.\, (\exists y.\, x = 2y) \wedge \exists z.\, x = 3z \\
\equiv\ & \exists x.\, 2 \mid x \wedge 3 \mid x \\
\equiv\ & \exists x.\, 6 \mid x \\
\equiv\ & \bigvee_{i=1}^{6} 6 \mid i \\
\equiv\ & \mathit{True}
\end{aligned}
$$

◊

Exercise 2 (Satisfiability algorithm for Presburger arithmetic)**.** Consider the formula $F(x)$ given by

$$F(x) = \bigwedge_{i=1} a_i < x \wedge \bigwedge_{j=1} x < b_j \wedge \bigwedge_{i=1} K_i | (x + t_i).$$

Note that the terms a_i, b_j, t_i may in general contain variables other than x.

1. First, assuming that all a_i, b_j, t_i are integer constants, give an algorithm that, given any formula of the form above, returns:

   - a value for x such that $F(x)$ holds, if one exists, and
   - "UNSAT" if no such value exists

   Solution: *Following the reasoning used in QE:*

   $$F(x) \equiv \bigvee_{i=1}^{\mathrm{lcm}(\{K_i\}_i)} F(\max_i a_i + i)$$

   A possible algorithm for satisfiability checks whether any of the disjuncts above (which are closed formulas) hold. If yes, it returns the argument of said disjunct. If none of the disjuncts hold, it returns "UNSAT". ◊

2. In the case that a_i, b_j, t_i may contain other variables, give a recursive algorithm that returns:

   - an assignment to each variable (including x) such that $F(x)$ holds, if one exists
   - "UNSAT" if no such assignment exists.

Solution: *Similar to the reasoning in the previous part, we can extract this information from the quantifier elimination procedure. Assuming the variables are $\bar{x} = (x, x_1, \dots, x_k)$, quantifier elimination for $\exists \bar{x}.\ F(x)$ gives us a formula of the form:*

$$\bigvee_{i=1}^{N} \bigvee_{i_1=1}^{N_1} \cdots \bigvee_{i_k=1}^{N_k} \varphi$$

with one disjunct per formula, in the (reversed) order of elimination. During the elimination, for each variable, we can note the value of the associated term $t_j = \max_i a_i$ used to eliminate it.

Finally, we can try (e.g. depth-first, backtracking) all possible values for $i, i_1, \dots, i_k$ to find a disjunct that holds. The associated values for $x, x_1, \dots, x_k$ can be computed from the recorded terms as $x_j := t_j + i_j$ for each j as before (and similarly for x). ♢

Exercise 3 (Quantifier elimination for rationals)**.** In this exercise we will devise a quantifier elimination method for rational numbers. We consider formulas over the signature $(\mathbb{Q}, <, \leq, =, +, -)$, i.e. with constant symbols among $\mathbb{Q}$, interpreted over the standard structure of rational numbers.

1. Show that for any formula F, there exists a formula F_1 such that

$$F \iff Q_1 x_1, \dots, Q_n x_n, Q_{n+1} y. F_1$$

where each Q_i is either $\exists$ or $\neg\exists$, i.e. existential quantifiers that can be separated by negations and where F_1 is built only from $(\wedge, \vee, \mathbb{Q}, <, =, +, -, k \cdot \_)$. In particular, it is quantifier-free and contains no negations.

Solution: *We can construct such a formula by transforming F by:*

(a) pushing all the quantifiers to the beginning of the formula (prenex normalization),

(b) changing $\forall x_i.\, \varphi$ into $\neg\exists x_i.\, \neg\varphi$,

(c) pushing negations to atomic subformulas, applying De Morgan's laws and converting $\neg\, a < b$ into $b \leq a$, $\neg\, a \leq b$ into $b < a$ and $\neg\, a = b$ into $b < a \vee a < b$, and

(d) substituting all occurrences of $a \leq b$ by $a < b \vee a = b$.

♢

2. Do we need to add a divisibility relation as in the Presburger arithmetic case? Why, or why not?

   Solution: *No, because in this case, variables are rationals, so unlike integers, they can be divided arbitrarily.* ♢

3. Show that there exist a formula F_2 such that $F_1 \iff F_2$ and every atom of F_2 containing y is of the form:

   $$t < y \quad \text{or} \quad y < t \quad \text{or} \quad t = y$$

   for some term t.

   Solution: *In each atom of F_1 containing y, we can move around all the variables except y in order to end up with atoms of the form $k \cdot y < t$, $t < k \cdot y$ or $t = k \cdot y$ for some term t and factor k. By multiplying both sides by $\frac{1}{k}$ we obtain the desired result.* ♢

4. Show that there exists a quantifier-free formula F_3 such that

   $$(\exists y. F_2) \iff F_3$$

   Solution: *There are several ways to tackle this problem. We keep the presentation here similar to the Presburger case to the extent possible.*

   We start by converting F_2 into DNF and distribute the existential quantifier over the disjunction afterwards. Therefore, we end up with subformulas of the form:

   $$\exists y.\, \varphi \wedge \bigwedge_{i \in I} t_i < y \wedge \bigwedge_{j \in J} y < t_j \wedge \bigwedge_{k \in K} t_k = y$$

   where φ is a quantifier-free formula where y does not appear, and I, J, K are index sets.

   - *If $K \neq \emptyset$: replace every occurrence of y by one of the t_k in the formula and remove the quantifier.*
   - *If $K = \emptyset$: the formula is equivalent to*

     $$\varphi \wedge \bigwedge_{i \in I} \bigwedge_{j \in J} t_i < t_j$$

     which is quantifier-free.

   ♢

Exercise 4 (Presburger Arithmetic without divisibility)**.** Following the steps below, show that Presburger Arithmetic without the divisibility predicate does *not* admit quantifier elimination.

1. For a formula $F(y)$ with one free variable y, define the set S_F as

   $$S_F = \{n \in \mathbb{N} \mid F(n)\}$$

   i.e. the set of positive integers that satisfy F.

   Show that there exists a formula $F(y)$ such that S_F is neither finite nor co-finite in $\mathbb{N}$ (i.e. S_F and $\mathbb{N} \setminus S_F$ are both infinite).

   Solution: *Consider $F(y) := \exists x. \, x + x = y$. Then, S_F is the set of even numbers, which is neither finite nor co-finite in $\mathbb{N}$.* $\Diamond$

2. Show that for any *quantifier-free* formula $G(y)$, S_G is either finite or co-finite in $\mathbb{N}$.

   Solution: *We can transform the formula $G(y)$ into an equivalent one by:*

   - *rewriting $t = s$ as $t - 1 < s < t + 1$ for all terms t and s,*
   - *rewriting constant relations (e.g. $3 < 5$) as either True or False,*
   - *rewriting to negation normal form, and*
   - *rewriting to disjunctive normal form.*

   After transformation, any quantifier-free formula $G(y)$ is of the form:

   $$\bigvee_i F_i(y)$$

   where each $F_i(y)$ is of the form:

   $$\bigwedge_j a_j < n_j \cdot y \wedge \bigwedge_k n'_k \cdot y < b_k$$

   Note that finite and cofinite sets are closed under finite unions. So, if each S_{F_i} is either finite or cofinite, then S_G is finite or cofinite as well. To clarify, this refers to each S_{F_i} being either finite or cofinite, not that they are all finite or all cofinite.

   We show that each S_{F_i} must be either finite or cofinite. The required result follows.

   Consider one such $F_i(y)$:

   $$\bigwedge_j a_j < n_j \cdot y \wedge \bigwedge_k n'_k \cdot y < b_k$$

This is equivalent to:

$$\bigwedge_j a_j \cdot \left(\frac{N}{n_j}\right) < N \cdot y \wedge \bigwedge_k N \cdot y < b_k \cdot \left(\frac{N}{n'_k}\right)$$

where $N = \mathrm{lcm}(\{n_j\}_j, \{n'_k\}_k)$.

We consider three cases:

- $\{a_i\}_i = \emptyset$*: there are no lower bounds on* y*, and we can merge the upper bounds to get*

  $$F_i \equiv N \cdot y < \min_k \left(b_k \cdot \left(\frac{N}{n'_k}\right)\right)$$

  and

  $$S_{F_i} = \left\{ y \in \mathbb{N} \,\middle|\, y \;<\; \left\lfloor \frac{\min_k \left(b_k \cdot \left(\frac{N}{n'_k}\right)\right)}{N} \right\rfloor \right\}$$

  which is a finite set.

- $\{b_j\}_j = \emptyset$*: there are no upper bounds on* y*, and we can merge the lower bounds to get*

  $$F_i \equiv \max_j \left(a_j \cdot \left(\frac{N}{n_j}\right)\right) < N \cdot y$$

  $$S_{F_i} = \left\{ y \in \mathbb{N} \,\middle|\, y \;>\; \left\lceil \frac{\max_j \left(a_j \cdot \left(\frac{N}{n_j}\right)\right)}{N} \right\rceil \right\}$$

  which is a cofinite set.

- *both* $\{a_i\}_i$ *and* $\{b_j\}_j$ *are non-empty: in this case,* S_{F_i} *is the intersection of the two sets defined above. The intersection of a finite and a cofinite set is finite, so* S_{F_i} *is finite.*

Thus we have shown that each S_{F_i} *is either finite or cofinite, and therefore* S_G *is either finite or cofinite as well.*

$\Diamond$

3. Using the results from the previous parts, show that Presburger Arithmetic without the divisibility predicate does not admit quantifier elimination.

   Solution: *Suppose Presburger arithmetic without the divisibility predicate admits quantifier elimination. Then, for any formula $F(y)$, there exists a quantifier-free formula $G(y)$ such that $F(y) \iff G(y)$. Therefore, $S_F = S_G$. But from the previous parts, we know that there exists a formula $F(y)$ such that S_F is neither finite nor cofinite, while for any quantifier-free formula $G(y)$, S_G is either finite or cofinite. This is a contradiction.* $\Diamond$

Exercise 5 (Structure of sets)**.** Consider the structure $(\mathcal{P}(\mathbb{N}), \subseteq, = \cap, \cup, \cdot^{\complement})$ whose base set is the set of all sets of natural numbers and where $\cdot^{\complement}$ denotes set complementation with respect to the universe.

Is it possible to eliminate quantifiers from arbitrary first-order formulas on this structure? For example, $\exists B. A \subseteq B \land B \subseteq C$ is equivalent to $A \subseteq C$.

Either:

1. Show that this structure admits quantifier elimination by giving a procedure to compute an equivalent quantifier-free formula from any given formula, **or**

2. Produce a quantified first-order logic formula that has no equivalent formula without quantifiers, and prove this property.

Solution:

Quantifier elimination is not possible as there exist formulas that have no quantifier-free equivalent. For example, consider the formula

$$\exists Y.\, X \cap Y^c \neq \emptyset \land X \cap Y \neq \emptyset$$

which is satisfied for all sets X containing at least two elements. Or equivalently that it is not equal to the empty set or a singleton.

Suppose such a quantifier-free formula $F(X)$ exists. Assume that this formula is in disjunctive normal form. Choose any disjunct D of this formula. We argue that this disjunct (a conjunction of literals) cannot distinguish between sets of cardinalities one and two.

First, consider the set of terms that can be built using just a variable X. Inductively, we can show that any such term is equivalent by the definitions of set operations to one of the following:

$$\emptyset, \mathbb{N}, X, X^c$$

Second, atomic formulas can only express that two such terms are equal or one is a subset of the other. Most are simple tautologies (e.g. $X \subseteq \mathbb{N}$) or fallacies (e.g. $\emptyset = \mathbb{N}$). The only non-trivial cases are:

$$X = \emptyset, X = \mathbb{N}$$

while all others, e.g. $X \subseteq X^{\complement} (\equiv X = \emptyset)$ can be reduced to these.

It is easy to see that no conjunction D of literals as above can distinguish between sets of cardinality one and two. Therefore, no disjunction $F(X)$ of such conjunctions can do so either. This contradicts our assumption that $F(X)$ is equivalent to the original formula.

$\Diamond$

Exercise 6 (A rational arithmetic formula)**.** Consider the following formula $G(x, z)$ where the variables range over rational numbers $\mathbb{Q}$:

$$\forall y.((x < y \land y < z) \longrightarrow \forall u.(x \neq u + u + u))$$

Find a quantifier-free formula equivalent to $G(x, z)$.

Solution:

$$\begin{aligned}
&\forall y.((x < y \land y < z) \longrightarrow \forall u.(x \neq u + u + u)) \\
\equiv\ &\neg\exists y.(x < y \land y < z \land \exists u.(x = u + u + u)) \\
\equiv\ &\neg\exists y.(x < y \land y < z \land \mathit{True}) \\
\equiv\ &\neg\exists y.(x < y \land y < z) \\
\equiv\ &\neg(x < z)
\end{aligned}$$

$\Diamond$